

# Transient gain with adaptation frozen vs active: what the first measurement says

2026-09-13, R5611351, Claude Code session 90c5825b-..., repo FractionalReservoir, branch main. Sections 1-6 were written after the fast smoke (HEAD e0810e4); section 7 adds the medium and 3-s-horizon runs the same evening. Data: data/transient\_gain\_smoke (fast mode:  $n = 500$ , one seed,  $T = 20$  s, six regular states per regime, horizon 1 s, 11 min). Figures: figs/transient\_gain\_smoke. Code: run\_transient\_gain, SRNNCellTypePairs.transient\_gain, fig\_transient\_gain, fig\_transient\_gain\_excursions. Background: Non\_normal\_amplification.md (the measure and why the dendritic block), Session\_2026-09-12-...\_handoff.md (the discussion that led here).

## 1. What was measured

At a sampled state  $S_0$  of the noisy trajectory, propagate the  $n$  dendritic unit vectors through the tangent equation and read the x-in / x-out block of the propagator every 20 ms:

$$G(t) = \| P_x \Phi(t) P_x^T \|_2 \quad (\text{worst case over dendritic directions})$$

plus, from the same  $n \times n$  block, the noise-average gain  $\|\cdot\|_F/\sqrt{n}$ , the gain along the E/I difference mode, the E/I sum mode and the leading Lyapunov direction at that state, and the top right singular vector at the peak (the optimal input pattern). Three propagators  $\Phi$  per state:

| name                   | $\Phi$                                            | what it is                                                                                                                |
|------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| J <sub>xx</sub> frozen | $e^{\{J_{xx} t\}}$                                | the dendritic block fixed at $S_0$ : the conventional rate-network linearisation, adaptation and depression held constant |
| J frozen               | $e^{\{J t\}}$                                     | the full Jacobian fixed at $S_0$ : a, b, g respond linearly, the state does not move                                      |
| active                 | $\Phi$ from $J(S(t))$ along the stored trajectory | what a tangent perturbation actually does                                                                                 |

Adjacent pairs isolate one effect each: J<sub>xx</sub> frozen → J frozen is adaptation's linear feedback at a fixed operating point; J frozen → active is nonstationarity (the state moves).  $G(0) = 1$  by construction; the propagator is verified against expm to 2e-6 on small networks.

## 2. Numbers (1-s horizon, medians over six states)

| regime                         | $\lambda_1$ | J <sub>xx</sub> frozen | J frozen | active | $e^{\{\lambda_1 \cdot 1 \text{ s}\}}$ |
|--------------------------------|-------------|------------------------|----------|--------|---------------------------------------|
| no adaptation                  | +4.0        | 47 000                 | 47 000   | 1 200  | 55                                    |
| single-timescale (sfa1_std1)   | +0.44       | 11 000                 | 143      | 30     | 1.6                                   |
| multiple-timescale (sfa3_std2) | -0.11       | 39                     | 6.7      | 8.6    | 0.9                                   |

The same numbers as ratios, which is how to read them:

| regime             | J <sub>xx</sub> frozen / J frozen | J frozen / active | active / $e^{\{\lambda_1 t\}}$ |
|--------------------|-----------------------------------|-------------------|--------------------------------|
| no adaptation      | 1 (no a, b states: $N = n$ )      | 39                | $\approx 22$                   |
| single-timescale   | 78                                | 4.8               | $\approx 20$                   |
| multiple-timescale | 5.8                               | 0.78              | $\approx 10$                   |

Direction readings (active propagator, at the peak): the optimal input pattern has a participation ratio of  $\sim 60$  of 500 neurons, E fraction  $\sim 0.5$ , and  $|\cos| \leq 0.15$  with the leading Lyapunov direction, in every regime. The E/I difference

mode is the best of the named directions in every regime; the E/I sum mode is the worst and falls below 1 once adaptation is on; the noise-average gain is  $\sim 1$  in the two adapted regimes.

Peak times: the two frozen propagators and the active one in the two unstable regimes peak AT THE HORIZON ( $t_{\text{peak}} = 1.0$  s), i.e. they are still growing. Only the multiple-timescale active curve has a genuine peak (0.18 s) followed by decay.

### 3. How to understand it

#### 3.1 The frozen picture is a runaway, not a transient, wherever $\alpha(J_{xx}) > 0$

In the no- and single-timescale regimes the dendritic block at a typical state has spectral abscissa  $\alpha \approx +7$  and  $+6$   $s^{-1}$  (fig\_transient\_amplification). Its propagator  $e^{J_{xx} t}$  therefore grows like  $e^{\alpha t}$  times a non-normal prefactor without bound; the “peak gain” is just the gain at the horizon. The  $\omega - \alpha$  margin of the earlier figure is the initial slope of that curve. So the static rate-network view does not describe a transient at all in those regimes: it describes exponential divergence at a frozen operating point that the real system never sits at. Comparisons involving the frozen propagators must be made at a fixed  $t$ , not at “the peak”.

#### 3.2 Adaptation’s linear feedback at the operating point is large

Holding the state fixed but letting  $a$  and  $b$  respond ( $J$  frozen) cuts the 1-s gain by  $78\times$  in the single-timescale regime and  $5.8\times$  in the multiple-timescale one. Mechanism: a dendritic perturbation  $\delta x$  raises  $\delta r$ , which raises  $\delta a$  on  $\tau_a$  and depletes  $\delta b$  on  $\tau_{\text{rec}}/\tau_{\text{rel}}$ , and the effective input  $x - c \cdot \Sigma a$  and the synaptic output  $r \cdot \Pi b$  both drop. This is the effect the instantaneous  $\omega$  cannot see (it is  $t = 0^+$ ), and it is the cleanest statement that adaptation belongs in the conceptual model of the recurrent dynamics: same weights, same state, two orders of magnitude less amplification in a second.

Why the reduction is smaller in the multiple-timescale regime ( $5.8\times$  vs  $78\times$ ): its  $J_{xx}$  frozen gain is already small (39 vs 11 000) because the operating point differs – STD has depressed  $W_{\text{eff}}$  and SFA has moved neurons to a shallower part of  $\phi$ , the  $\omega = 46$  vs  $74$  story – so there is less left for the feedback to remove within 1 s. The 10-s SFA rung has not acted within this horizon at all.

#### 3.3 Nonstationarity is a second, separate reduction, present even without adaptation

Without adaptation the two frozen propagators are identical (a sanity check of the code:  $N = n$ ), yet the active gain is  $39\times$  smaller than the frozen one. The linearisation at any single point of a chaotic trajectory has  $\alpha \approx +7$ , but the trajectory diverges only at  $\lambda_1 = +4$ : the unstable directions rotate as the state moves, and no direction stays aligned with the local worst case for long. Freezing the Jacobian overstates amplification in every regime, adapted or not. In the single-timescale regime the  $J$ -frozen curve’s late log-slope is about  $+3$   $s^{-1}$  against  $\lambda_1 = +0.44$ : the trajectory is more stable than any of the states along it, which is what “stabilisation through motion” looks like when adaptation keeps moving the operating point.

#### 3.4 What actually amplifies: 10-20 $\times$ along a specific pattern

The honest quantity is the active gain divided by  $e^{\lambda_1 t}$ : the transient amplification over and above asymptotic divergence or decay. It is 22, 20 and 10 in the three regimes. That is the Murphy-Miller / Hennequin balanced-amplification quantity for this network, and adaptation reduces it only by about  $2\times$ ; the four-orders-of-magnitude spread in the frozen column is mostly the change in  $\lambda_1$  ( $+4 \rightarrow -0.1$ ) and in the operating point, not a change in non-normality. In the stable regime the whole  $8.6\times$  is a transient: peak at 0.18 s, then decay, the textbook shape. In the other two regimes the transient rides on exponential divergence and never peaks within a second.

#### 3.5 The amplifying pattern is not the chaotic direction

The optimal input pattern is a  $\sim 60$ -neuron E/I-imbalanced pattern with  $|\cos| \leq 0.15$  to the leading Lyapunov direction (whose  $x$  part we compared; the full leading vector lives  $\sim 80\%$  in the SFA states in the adapted regimes). Two different mechanisms, then: chaos expands along a slow, adaptation-dominated direction; transient amplification is a fast dendritic E/I pattern. The E/I difference mode being the best named direction everywhere confirms the balanced-amplification prediction; the E/I sum mode being damped ( $< 1$ ) in the adapted regimes says a uniform

push engages every neuron’s adaptation and is cancelled – relevant to the stimulation idea (Next 3), where the prior was that the sum mode is the direction that engages adaptation without a large transient.

The noise-average gain  $\sim 1$  in the adapted regimes means the network’s own additive noise is not amplified on average; only structured perturbations are, 10-30 $\times$  along the optimal pattern. Adaptation makes the network selectively sensitive rather than globally quiet.

### 3.6 Active can exceed J frozen (the stable regime: 8.6 vs 6.7)

The plan expected the frozen gains to bound the active one; they do not. The local Lyapunov rate fluctuates around  $\lambda_1$  (positive about a third of the time in this regime, in excursions of a few hundred ms), and a moving state passes through stretches with more expansion than the sampled point had; time-varying linear systems can amplify more than any of their frozen coefficients allow. The same ordering appeared on the 40- and 60-neuron test networks. With six states from one seed the 8.6 vs 6.7 difference is inside the state-to-state spread, so it is a caution (“do not call the frozen gains bounds; report all three”), not yet a result.

## 4. What this means for the paper

- The conventional rate-network linearisation ( $J_{xx}$ ) overstates amplification by orders of magnitude in the adapted regimes, and by 40 $\times$  even without adaptation. Say so, with the three-curve figure.
- Adaptation’s contribution at the operating point is the  $J_{xx}$ -frozen vs J-frozen ratio (78 $\times$  and 5.8 $\times$  at 1 s). Its contribution to the asymptotic rate is  $\lambda_1$  (+4  $\rightarrow$  +0.44  $\rightarrow$  -0.11). Its contribution to the non-normal prefactor is modest (22  $\rightarrow$  20  $\rightarrow$  10). Keep these three apart; the frozen picture conflates them.
- The number to quote for “how much does a dendritic perturbation grow before adaptation reverses it” is the active peak in the stable regime: 8.6 $\times$  at 0.18 s (one seed; get the medium-mode value). In the unstable regimes quote the gain at a fixed t together with  $e^{\{\lambda_1 t\}}$ .
- The amplifying pattern is an E/I difference pattern over  $\sim 60$  neurons, distinct from the Lyapunov direction; the E/I sum pattern is damped.
- Keep the  $\omega - \alpha$  figure as the conventional view, captioned as the initial slope of the frozen curve.

## 5. Caveats

One seed, six states, 1-s horizon. Tangent-space linearisation along a noisy trajectory: G is an upper envelope of real amplification (saturation ignored). The horizon is too short for the 10-s SFA rung. The excursion sampling (Next 2) found only 0-2 onset / quiet states per regime in 20 s; the onset-vs-quiet contrast is untested.

## 6. What to do next

1. Run the stage at medium ( $T = 40$  s, 2 seeds, 12 regular + up to 8 onset
  - 8 quiet, horizon 2 s,  $\sim 1$  h): `run_transient_gain('run_mode','medium','out_dir', full-file('data','topk_med','transient_gain'))`, then the two figures with `run_dir = data/topk_med`. This gives the resolved peak in the stable regime, the seed spread, and the first real onset-vs-quiet contrast.
2. Record the spectral abscissa of the FULL J at every sampled state (a few eigs per state, or the late log-slope of the J-frozen curve) next to  $\lambda_1$ : “the trajectory is more stable than any of its states” is a claim worth a number.
3. A 3-s horizon on fewer states, so the 10-s rung begins to act and the single-timescale curve can be seen to peak or not.
4. Next 3 (stimulation directions): the E/I sum mode’s damping supports the prior; the intermittent-but-stable networks are listed by `find_intermittent_stable` (multiple-timescale regime at  $\mu_{IE\_relative}$  2.06 or  $\mu_{EE\_relative}$  13.4-15.7; single-timescale at  $n = 190$ ). ## 7. Medium run and 3-s horizon (2026-09-13 evening): the picture holds, with numbers

Two further runs after the smoke, both with the stage now recording  $\alpha(J_{xx})$ ,  $\omega(J_{xx})$  and  $\alpha(J)$  at every sampled state (commit f18d83f):

- **Medium** (data/topk\_med/transient\_gain, figs/topk\_med):  $T = 40$  s, 2 seeds, 12 regular + up to 8 onset + 8 quiet states per seed, horizon 2 s, 51 min. 24 regular states per regime.

- **3-s horizon** (data/transient\_gain\_h3, figs/transient\_gain\_h3):  $T = 40$  s, 1 seed, 4 regular states per regime, horizon 3 s, 21 min.

(The first medium attempt died after 51 min at the assembly step – the per-seed sample arrays have different lengths and were concatenated horizontally; fixed in 43ef72e, which also saves the raw trials before assembly and makes the stage test run two seeds.)

### 7.1 The operating point vs the trajectory (medium, medians over 24-40 states)

| regime             | $\lambda_1$ (trajectory) | $\alpha(J_{xx})$ at state | $\omega(J_{xx})$ at state | $\alpha(J)$ at state                            |
|--------------------|--------------------------|---------------------------|---------------------------|-------------------------------------------------|
| no adaptation      | +3.35 [2.63, 4.08]       | +6.0 [1.9, 10.7]          | 80                        | +6.0 (= $\alpha(J_{xx})$ , $N = n$ )            |
| single-timescale   | +0.48                    | +5.9 [2.7, 10.0]          | 80                        | +1.8 [-0.3, +4.7]                               |
| multiple-timescale | -0.11                    | +0.7 [-0.8, +3.9]         | 50                        | +0.5 [-0.2, +1.9] (21 of 31 states unconverged) |

Three facts in one table. (i) Adaptation's linear feedback at the operating point lowers the frozen growth rate from +5.9 to +1.8  $s^{-1}$  (single) and from +0.7 to +0.5 (multiple): that is what the  $J_{xx}$ -frozen vs  $J$ -frozen gap is made of. (ii) **The trajectory is more stable than any of its states:**  $\alpha(J)$  at a typical state is +1.8 where  $\lambda_1$  is +0.48, and +0.5 where  $\lambda_1$  is -0.11. Without adaptation the same gap (+6.0 vs +3.35) exists and is pure nonstationarity; with adaptation the operating point keeps moving away from its own unstable directions. (iii)  $\omega - \alpha$  is  $\sim 75 s^{-1}$  in every regime, i.e. the instantaneous non-normal margin barely changes with adaptation even though the finite-time gain changes by orders of magnitude – the margin is the wrong number to summarise the effect with.

Caveat on  $\alpha(J)$ : `eigs(J, 6, 'largestreal')` on the sparse  $4000 \times 4000$  often converges only some of the six; the largest converged real part is taken, and in the stable regime two thirds of the states returned none. Treat it as an estimate (the  $J$ -frozen curve's late log-slope agrees:  $\sim +1.5$  to  $+2 s^{-1}$  in the single-timescale regime).

### 7.2 Gains (medium, horizon 2 s, 24 regular states, median [min, max])

| regime             | $J_{xx}$ frozen                                 | $J$ frozen                    | active                           | active $t_{peak}$ | active / $e^{\{2\lambda_1\}}$ |
|--------------------|-------------------------------------------------|-------------------------------|----------------------------------|-------------------|-------------------------------|
| no adaptation      | $3 \times 10^6$ [ $5 \times 10^2$ , $10^{10}$ ] | same                          | 11 600 [650, $1.4 \times 10^6$ ] | 2.0 (horizon)     | 14                            |
| single-timescale   | $10^6$ [ $10^3$ , $4 \times 10^9$ ]             | 119 [ $4$ , $6 \times 10^4$ ] | 48 [23, 410]                     | 1.94 [0.8, 2.0]   | 18                            |
| multiple-timescale | 22 [ $5$ , $2 \times 10^4$ ]                    | 4.8 [3.3, 69]                 | 5.0 [3.6, 12.4]                  | 0.14 [0.10, 1.4]  | 6                             |

With the 3-s horizon (one seed, 4 states), the active median curve reads:

| regime             | peak                   | at 1 s | 2 s    | 3 s             | noise-average at 3 s |
|--------------------|------------------------|--------|--------|-----------------|----------------------|
| no adaptation      | $3 \times 10^6$ at 3 s | 820    | 23 000 | $3 \times 10^6$ | $1.4 \times 10^5$    |
| single-timescale   | 155 at 2.7 s           | 37     | 55     | 122             | 6                    |
| multiple-timescale | 5.1 at 0.14 s          | 2.5    | 1.0    | 0.50            | 0.03                 |

So: the frozen numbers past  $\sim 1$  s are astronomically large and meaningless where  $\alpha > 0$  (the range spans seven decades because  $e^{\{\alpha t\}}$  with  $\alpha \in [2, 10]$  does); the active gain is what a perturbation does. In the **stable regime a dendritic perturbation is amplified 5 $\times$  within 140 ms, is back to its initial size by 2 s, and is at half by 3 s**; the isotropic (noise-average) response is at 3% by then. The 10-s SFA rung produces no second rise within 3 s (there is a small shoulder near 1 s in the medium figure). In the single-timescale regime the active gain never peaks: it rides on  $\lambda_1 = +0.48$  with a transient prefactor of  $\sim 20$ -50 that is still growing at 3 s; the  $J$ -frozen curve, which the smoke had at 143 at 1 s, is at  $3.6 \times 10^6$  by 3 s – the frozen operating point is unstable at +1.8  $s^{-1}$  and the trajectory is not.

The transient prefactor active /  $e^{\{\lambda_1 t\}}$  at 2 s is 14 / 18 / 6: the same  $\sim 2\times$  reduction by adaptation as the smoke showed, against four decades in the frozen column. The direction readings are unchanged: participation  $\sim 20$  (no adaptation) to  $\sim 70$  (adapted) neurons, E fraction  $\sim 0.5$ ,  $|\cos|$  with the leading Lyapunov direction 0.04-0.06 in every regime and every propagator; E/I difference the best named direction, E/I sum below 1 when adapted; noise-average  $\approx 1$  (single) and  $< 1$  (multiple).

### 7.3 Onsets vs quiet states (Next 2): a real but modest effect, not along the Lyapunov direction

Found in  $2 \times 40$  s: 3 / 19 / 4 onsets and 0 / 0 / 3 quiet stretches ( $\geq 1$  s of negative local rate) for no / single / multiple-timescale adaptation. A 1-s quiet stretch does not occur in the single-timescale regime (the local rate is positive about half the time), so the planned onset-vs-quiet contrast is empty there. Post hoc, the regular states split by the local rate at the sample (stored) give the contrast instead:

| regime                            | onset: n, active<br>G_max (2 s) | regular with local<br>rate < 0: n, G_max | rank-sum p       | G along $v_{\text{Lyap}}$ at<br>the peak, onset vs<br>contracting |
|-----------------------------------|---------------------------------|------------------------------------------|------------------|-------------------------------------------------------------------|
| single-timescale                  | 16, 62 [27, 200]                | 13, 42                                   | 0.09             | 4.4 vs 3.1                                                        |
| multiple-timescale                | 4, 8.2 [6.2, 11.0]              | 24, 5.0                                  | 0.02             | 0.87 vs 0.72                                                      |
| multiple-timescale,<br>1-s quiets | –                               | 3, 5.2 [4.6, 6.7]                        | 0.11 (vs onsets) | –                                                                 |

Onset states carry  $\sim 1.5\times$  more worst-case gain than contracting states, significant in the stable regime with 4 vs 24 states and marginal in the single-timescale one. But the optimal direction at an onset has  $|\cos|$  0.04-0.05 with the leading Lyapunov direction, the same as anywhere else, and the gain along the Lyapunov direction itself is small (4.4 vs 62 worst case; 0.87, i.e. contraction, in the stable regime). So the excursions are NOT the network recruiting its worst-case non-normal mode. The state at an onset is somewhat more amplifying in every direction (a higher-gain operating point: more neurons on the steep part of  $\phi$ , less depressed synapses), and the divergence itself happens along a slow, adaptation-dominated direction that a dendritic perturbation barely projects onto. The mechanistic account of intermittency is therefore “operating-point excursions”, not “transient amplification along the unstable direction”.

### 7.4 Updated reading for the paper

Everything in §4 stands; two sharpenings. First, the number to quote in the stable regime is now 5.0 [3.6, 12.4] at 0.14 s over 24 states and 2 seeds, and “gone by 2 s”. Second, the operating-point table (§7.1) is the cleanest single statement of adaptation’s role: at a typical state the recurrent block alone is unstable at  $+6 \text{ s}^{-1}$ , adaptation’s feedback at that state brings it to  $+1.8$  or  $+0.5$ , and the trajectory’s own rate is  $+0.48$  or  $-0.11$  – each step is a mechanism the frozen picture lacks.

### 7.5 What is still open

- `fig_transient_gain_excursions` should draw the post-hoc split (onset vs regular-contracting) as well as onset vs 1-s quiets, and the quiet threshold should be a `cfg` option (0.5 s would exist in the single-timescale regime). Not done; the numbers above came from a console analysis of the saved samples.
- $\alpha(J)$  via `eigs` converges poorly on the stable regime’s Jacobian; a shift-invert or the late log-slope of the J-frozen curve would be more reliable.
- Only two seeds; production mode (3 seeds, 3-s horizon, 20 regular states) is the version for the manuscript.
- Next 3 (stimulation directions): the E/I sum mode is damped in the adapted regimes and the amplifying pattern is a  $\sim 70$ -neuron E/I-difference pattern, which supports the prior that a uniform push engages adaptation without a large transient. The candidate networks are in §6 of `Non_normal_amplification.md` / `find_intermittent_stable`.